

Отчёт по лабораторной работе № 6

Название работы: Интерактивный rebase, cherry-pick, теги, reflog

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1. Выполнение шагов лабораторной работы

Отметьте галочками выполненные шаги. В столбце «Команды / вывод» напишите выполненные команды и полученный вывод (поле автоматически растягивается по содержимому).

Шаг	Описание	Вып.	Команды / вывод
1	Настроить редактор Git на nano (git config --global core.editor 'nano')	☒	PS C:\Users\HOME\lab5_gitflow> git config --global core.editor "nano" PS C:\Users\HOME\lab5_gitflow> git config --global core.editor nano
2	Создать репозиторий и сделать 5 коммитов (Step 1..5)	☒	PS C:\Users\HOME\lab5_gitflow> mkdir lab6_advanced >> cd lab6_advanced >> git init Каталог: C:\Users\HOME\lab5_gitflow Mode LastWriteTime Length Name ---- ----- d----- 09.05.2026 0:45 lab6_advanced Initialized empty Git repository in C:/Users/HOME/lab5_gitflow/lab6_adva nced/.git/

Шаг	Описание	Вып.	Команды / вывод
			PS C:\Users\HOME\lab5_gitflow\lab6_advanced> echo "Commit 1" > f1.txt >> git add . >> git commit -m "Step 1" [master (root-commit) 59900ad] Step 1 1 file changed, 0 insertions(+), 0 deletions(-) create mode 100644 f1.txt PS C:\Users\HOME\lab5_gitflow\lab6_advanced> echo "Commit 2" > f2.txt >> git add . >> git commit -m "Step 2" [master 1d4b1c6] Step 2 1 file changed, 0 insertions(+), 0 deletions(-) create mode 100644 f2.txt PS C:\Users\HOME\lab5_gitflow\lab6_advanced> echo "Commit 3" > f3.txt >> git add . >> git commit -m "Step 3" [master 387a11a] Step 3 1 file changed, 0 insertions(+), 0 deletions(-) create mode 100644 f3.txt PS C:\Users\HOME\lab5_gitflow\lab6_advanced> echo "Commit 4" > f4.txt >> git add . >> git commit -m "Step 4" [master 1b9969f] Step 4 1 file changed, 0 insertions(+), 0 deletions(-) create mode 100644 f4.txt PS C:\Users\HOME\lab5_gitflow\lab6_advanced> >> echo "Commit 5" > f5.txt >> git add . >> git commit -m "Step 5" [master f4ec36d] Step 5 1 file changed, 0 insertions(+), 0 deletions(-)

Шаг	Описание	Вып.	Команды / вывод
			<pre> create mode 100644 f5.txt PS C:\Users\HOME\lab5_gitflow\lab6_advanced> git log --oneline f4ec36d (HEAD -> master) Step 5 1b9969f Step 4 387a11a Step 3 1d4b1c6 Step 2 59900ad Step 1 </pre>
3	Создать ветку feature-rebase, добавить три коммита (Feature A, Feature B, Fix typo)	☒	<pre> PS C:\Users\HOME\lab5_gitflow\lab6_advanced> git checkout -b feature-rebase Switched to a new branch 'feature-rebase' PS C:\Users\HOME\lab5_gitflow\lab6_advanced> echo "Feature A" > fa.txt >> git add . >> git commit -m "Feature A" [feature-rebase bef5074] Feature A 1 file changed, 0 insertions(+), 0 deletions(-) create mode 100644 fa.txt PS C:\Users\HOME\lab5_gitflow\lab6_advanced> echo "Feature B" > fb.txt >> git add . >> git commit -m "Feature B" [feature-rebase ad24474] Feature B 1 file changed, 0 insertions(+), 0 deletions(-) create mode 100644 fb.txt PS C:\Users\HOME\lab5_gitflow\lab6_advanced> echo "Fix" >> f3.txt >> git commit -am "Fix typo" [feature-rebase ab38b21] Fix typo 1 file changed, 0 insertions(+), 0 deletions(-) PS C:\Users\HOME\lab5_gitflow\lab6_advanced> git log --oneline ab38b21 (HEAD -> feature-rebase) Fix typo </pre>

Шаг	Описание	Вып.	Команды / вывод
			ad24474 Feature B bef5074 Feature A f4ec36d (master) Step 5 1b9969f Step 4 387a11a Step 3 1d4b1c6 Step 2 59900ad Step 1
4	Запустить git rebase -i HEAD~3 и изменить: squash для Feature B, reword для Fix typo	☒	GNU nano 9.0 C:/Users/HOME/lab5_gitflow/lab6_advanced/.git/COMMIT_EDITMSG # This is a combination of 2 commits. # This is the 1st commit message: Feature A # This is the commit message #2: Feature B # Please enter the commit message for your changes. Lines starting # with '#' will be ignored, and an empty message aborts the commit. # # Date: Sat May 9 00:46:46 2026 +0500 # # interactive rebase in progress; onto f4ec36d # Last commands done (2 commands done): # pick bef5074 # Feature A # squash ad24474 # Feature B # Next command to do (1 remaining command): # reword ab38b21 # Fix typo # You are currently rebasing branch 'feature-rebase' on 'f4ec36d'. # # Changes to be committed: # new file: fa.txt # new file: fb.txt

Шаг	Описание	Вып.	Команды / вывод
5	Сохранить изменения в редакторе (nano: Ctrl+O, Enter, Ctrl+X)	☒	<pre> GNU nano 9.0 C:/Users/HOME/lab5_gitflow/lab6_advanced/.git/COMMIT_EDITMSG # This is a combination of 2 commits. # This is the 1st commit message: Feature A # This is the commit message #2: Feature B # Please enter the commit message for your changes. Lines starting # with '#' will be ignored, and an empty message aborts the commit. # # Date: Sat May 9 00:46:46 2026 +0500 # # interactive rebase in progress; onto f4ec36d # Last commands done (2 commands done): # pick bef5074 # Feature A # squash ad24474 # Feature B # Next command to do (1 remaining command): # reword ab38b21 # Fix typo # You are currently rebasing branch 'feature-rebase' on 'f4ec36d'. # # Changes to be committed: # new file: fa.txt # new file: fb.txt </pre>
6	При squash объединить сообщения коммитов, при reword написать 'Fix critical typo in f3.txt'	☒	<pre> GNU nano 9.0 C:/Users/HOME/lab5_gitflow/lab6_advanced/.git/COMMIT_EDITMSG # This is a combination of 2 commits. # This is the 1st commit message: Feature A and B combined </pre>

Шаг	Описание	Вып.	Команды / вывод
			<pre># This is the commit message #2: # Please enter the commit message for your changes. Lines starting # with '#' will be ignored, and an empty message aborts the commit. # # Date: Sat May 9 00:46:46 2026 +0500 # # interactive rebase in progress; onto f4ec36d # Last commands done (2 commands done): # pick bef5074 # Feature A # squash ad24474 # Feature B # Next command to do (1 remaining command): # reword ab38b21 # Fix typo # You are currently rebasing branch 'feature-rebase' on 'f4ec36d'. # # Changes to be committed: # new file: fa.txt # new file: fb.txt # 3e31bca (HEAD -> feature-rebase) Fix critical typo in f3.txt 9d75554 Feature A and B combined f4ec36d (master) Step 5 1b9969f Step 4 387a11a Step 3 1d4b1c6 Step 2 59900ad Step 1</pre>
7	Проверить историю (git log --oneline) — коммитов стало меньше	☒	<pre>PS C:\Users\HOME\lab5_gitflow\lab6_advanced> git checkout master Switched to branch 'master' PS C:\Users\HOME\lab5_gitflow\lab6_advanced> git cherry-pick 3e31bca [master 9367b39] Fix critical typo in f3.txt Date: Sat May 9 00:46:52 2026 +0500</pre>

Шаг	Описание	Вып.	Команды / вывод
			1 file changed, 0 insertions(+), 0 deletions(-) PS C:\Users\HOME\lab5_gitflow\lab6_advanced> git log --oneline 9367b39 (HEAD -> master) Fix critical typo in f3.txt f4ec36d Step 5 1b9969f Step 4 387a11a Step 3 1d4b1c6 Step 2 59900ad Step 1
8	Переключиться в main и скопировать исправленный коммит через cherry-pick	☒	PS C:\Users\HOME\lab5_gitflow\lab6_advanced> git tag v1.0.0 PS C:\Users\HOME\lab5_gitflow\lab6_advanced> git tag -a v1.1.0 -m "Аннотированный тег" PS C:\Users\HOME\lab5_gitflow\lab6_advanced> git tag -n v1.0.0 Fix critical typo in f3.txt v1.1.0 Аннотированный тег
9	Создать лёгкий тег v1.0.0 и аннотированный тег v1.1.0	☒	PS C:\Users\HOME\lab5_gitflow\lab6_advanced> git checkout v1.0.0 Note: switching to 'v1.0.0'. You are in 'detached HEAD' state. You can look around, make experimental changes and commit them, and you can discard any commits you make in this state without impacting any branches by switching back to a branch. If you want to create a new branch to retain commits you create, you may do so (now or later) by using -c with the switch command. Example: git switch -c <new-branch-name>

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			Or undo this operation with: git switch - Turn off this advice by setting config variable advice.detachedHead to false HEAD is now at 9367b39 Fix critical typo in f3.txt PS C:\Users\HOME\lab5_gitflow\lab6_advanced> git checkout -b hotfix-from-tag Switched to a new branch 'hotfix-from-tag' PS C:\Users\HOME\lab5_gitflow\lab6_advanced> git branch feature-rebase * hotfix-from-tag master
10	Переключиться по тегу v1.0.0 (detached HEAD), создать от него ветку hotfix-from-tag	☒	PS C:\Users\HOME\lab5_gitflow\lab6_advanced> git checkout master Switched to branch 'master' PS C:\Users\HOME\lab5_gitflow\lab6_advanced> git reset --hard HEAD~3 HEAD is now at 387a11a Step 3 PS C:\Users\HOME\lab5_gitflow\lab6_advanced> git log --oneline 387a11a (HEAD -> master) Step 3 1d4b1c6 Step 2 59900ad Step 1 PS C:\Users\HOME\lab5_gitflow\lab6_advanced> git reflog 387a11a (HEAD -> master) HEAD@{0}: reset: moving to HEAD~3 9367b39 (tag: v1.1.0, tag: v1.0.0, hotfix-from-tag) HEAD@{1}: checkout: moving from hotfix-from-tag to master 9367b39 (tag: v1.1.0, tag: v1.0.0, hotfix-from-tag) HEAD@{2}: checkout: moving

Шаг	Описание	Вып.	Команды / вывод
			from 9367b39e8ff2b8e21bfa286ca3d548d3a3e4b6d4 to hotfix-from-tag 9367b39 (tag: v1.1.0, tag: v1.0.0, hotfix-from-tag) HEAD@{3}: checkout: moving from master to v1.0.0 9367b39 (tag: v1.1.0, tag: v1.0.0, hotfix-from-tag) HEAD@{4}: cherry-pick: Fix critical typo in f3.txt f4ec36d HEAD@{5}: checkout: moving from feature-rebase to master 3e31bca (feature-rebase) HEAD@{6}: rebase (finish): returning to refs/heads/feature-rebase 3e31bca (feature-rebase) HEAD@{7}: rebase (reword): Fix critical typo in f3.txt 6151791 HEAD@{8}: rebase (reword): Fix typo 9d75554 HEAD@{9}: rebase (squash): Feature A and B combined bef5074 HEAD@{10}: rebase (start): checkout HEAD~3 ab38b21 HEAD@{11}: commit: Fix typo ad24474 HEAD@{12}: commit: Feature B bef5074 HEAD@{13}: commit: Feature A f4ec36d HEAD@{14}: checkout: moving from master to feature-rebase f4ec36d HEAD@{15}: commit: Step 5 1b9969f HEAD@{16}: commit: Step 4 387a11a (HEAD -> master) HEAD@{17}: commit: Step 3 1d4b1c6 HEAD@{18}: commit: Step 2 59900ad HEAD@{19}: commit (initial): Step 1 PS C:\Users\HOME\lab5_gitflow\lab6_advanced> git reset --hard HEAD@{1} error: unknown switch `e' usage: git reset [--mixed --soft --hard --merge --keep] [-q] [<commit>] or: git reset [-q] [<tree-ish>] [--<paths-spec>... or: git reset [-q] [--paths-spec-from-file [--

Шаг	Описание	Вып.	Команды / вывод
			pathspec-file-nul]] [<tree-ish>] or: git reset --patch [<tree-ish>] [-- [<pathspec>...] or: DEPRECATED: git reset [-q] [-- stdin [-z]] [<tree-ish>] -q, --[no-]quiet be quiet, only report errors --no-refresh skip refreshing the index after reset --refresh opposite of --no-refresh --mixed reset HEAD and index --soft reset only HEAD --hard reset HEAD, index and working tree --merge reset HEAD, index and working tree --keep reset HEAD but keep local changes --[no-]recurse-submodules[=<reset>] control recursive updating of submodules -p, --[no-]patch select hunks interactively --[no-]auto-advance auto advance to the next file when selecting hunks interactively -U, --unified <n> generate diffs with <n> lines context --inter-hunk-context <n> show context between diff hunks up to the specified number of lines -N, --[no-]intent-to-add record only the fact that removed paths will be added later --[no-]pathspec-from-file <file> read pathspec from file --[no-]pathspec-file-nul with --pathspec-from-file, pathspec elements are separated with NUL character -z DEPRECATED (use --pathspec-file-nul instead): paths are

Шаг	Описание	Вып.	Команды / вывод
			separated with NUL character --[no-]stdin DEPRECATED (use --pathspec-from-file=- instead): read paths from <stdin> PS C:\Users\HOME\lab5_gitflow\lab6_adv anced> git log --oneline 387a11a (HEAD -> master) Step 3 1d4b1c6 Step 2 59900ad Step 1
11	Выполнить git reset --hard HEAD~3, затем восстановить потерянные коммиты через reflog	☒	PS C:\Users\HOME\lab5_gitflow\lab6_adv anced> git reset --hard 9367b39 HEAD is now at 9367b39 Fix critical typo in f3.txt PS C:\Users\HOME\lab5_gitflow\lab6_adv anced> git log --oneline 9367b39 (HEAD -> master, tag: v1.1.0, tag: v1.0.0, hotfix-from-tag) Fix critical typo in f3.txt f4ec36d Step 5 1b9969f Step 4 387a11a Step 3 1d4b1c6 Step 2 59900ad Step 1

2. Самостоятельное задание

Условие:

Создайте ветку messy, сделайте 4 коммита. С помощью git rebase -i HEAD~4 объедините первые два коммита и переименуйте третий. Перенесите один коммит из messy в main через cherry-pick. Создайте аннотированный тег exercise-v1.

Выполнение (ход действий и результат):

```
PS C:\Users\HOME\lab5_gitflow\lab6_advanced> git checkout -b messy
echo "One" > one.txt
git add .
git commit -m "One"
echo "Two" > two.txt
git add .
```

```
git commit -m "Two"
echo "Three" > three.txt
git add .
git commit -m "Three"
echo "Four" > four.txt
git add .
git commit -m "Four"
Switched to a new branch 'messy'
[messy 1f9edc7] One
1 file changed, 0 insertions(+), 0 deletions(-)
create mode 100644 one.txt
[messy c2139e1] Two
1 file changed, 0 insertions(+), 0 deletions(-)
create mode 100644 two.txt
[messy 1c47844] Three
1 file changed, 0 insertions(+), 0 deletions(-)
create mode 100644 three.txt
[messy 2454b99] Four
1 file changed, 0 insertions(+), 0 deletions(-)
create mode 100644 four.txt
PS C:\Users\HOME\lab5_gitflow\lab6_advanced> git rebase -i HEAD~4
GNU nano 9.0      C:/Users/HOME/lab5_gitflow/lab6_advanced/.git/rebase-merge/git-rebase-
todo      Modified
ipick 1f9edc7 # One
squash c2139e1 # Two
reword 1c47844 # Three
pick 2454b99 # Four
d1c32cc (HEAD -> messy) Four
bec9871 Third comit renamed
1e3ee4b One and Two combined
PS C:\Users\HOME\lab5_gitflow\lab6_advanced> git checkout master
Switched to branch 'master'
PS C:\Users\HOME\lab5_gitflow\lab6_advanced> git cherry-pick d1c32cc
[master 010b7fc] Four
Date: Sat May 9 00:59:32 2026 +0500
1 file changed, 0 insertions(+), 0 deletions(-)
create mode 100644 four.txt
PS C:\Users\HOME\lab5_gitflow\lab6_advanced> git tag -a exercise-v1 -m "Exercise version 1"
PS C:\Users\HOME\lab5_gitflow\lab6_advanced> git tag -n
exercise-v1      Exercise version 1
v1.0.0          Fix critical typo in f3.txt
v1.1.0          Аннотированный тег
```

3. Ответы на контрольные вопросы

1. Вопрос 1:

`git rebase` переписывает историю коммитов

если ветка уже отправлена на удалённый репозиторий то у других пользователей появятся конфликты и разные версии

2. Вопрос 2:

3. `git cherry-pick` переносит один конкретный коммит

4. `git merge` объединяет целую ветку со всей историей

5. Вопрос 3:

`git reflog`

`git reset --hard [ХЭШ]`

4. Дополнительные замечания (при необходимости)

Вопрос 4:

пример как установить nano

`git config --global core.editor "nano"`

Инструкция: Выберите номер лабораторной работы. Заполняйте поля — они автоматически растягиваются по высоте под текст. Для печати нажмите кнопку «Печать» — все рамки полей исчезнут, и отчёт будет выглядеть как обычный документ.